Analysis of Prompting Techniques in LLM Reasoning

GENAI EX-5.ipynb

AIM:

To Develop a Python program to analyze how different prompting strategies—Zero-shot, Few-shot, and Chain-of-Thought (CoT)—affect the reasoning capabilities of Large Language Models (LLMs). The program will also identify patterns in logical fallacies or hallucinations using benchmark datasets.

CODE:

```
# Install once
!pip install -q transformers google-generativeai sentencepiece

import re, time
import pandas as pd
from transformers import pipeline
import google.generativeai as genai
from getpass import getpass

# Gemini API
genai.configure(api_key=getpass("Enter Gemini API Key: "))

# Models
hf = pipeline("text-generation", model="distilgpt2")
gm = genai.GenerativeModel("gemini-2.5-flash")

# Sample Questions (YOU CONTROL INPUT HERE)
questions = [
   "If you have 5 apples and you buy 3 more, how many apples do you
have?",
   "A train travels 60 km in 1 hour. How far in 3 hours?",
   "What is 25 + 17?"
]

# Prompt styles
def zero(q): return f"Q: {q}\nA:"
def few(q): return f"Q: 2+3?\nA: 5\nQ: 10-4?\nA: 6\nQ: {q}\nA:"
```

```python
def cot(q): return f"Solve step by step.\nQ: {q}\nA:"

# Ask models
def ask_hf(p):
    try:
        return hf(p, max_new_tokens=40, do_sample=False,
pad_token_id=50256)[0]["generated_text"]
    except:
        return "HF Error"

def ask_gm(p):
    try:
        return gm.generate_content(p).text
    except:
        return "Quota exceeded"

# Extract numbers
def get_num(text):
    nums = re.findall(r"-?\d+\.?\d*", str(text))
    return nums[-1] if nums else None

# Run
results = []

for q in questions:
    for name, fn in [("Zero-shot", zero), ("Few-shot", few), ("CoT", cot)]:
        prompt = fn(q)

        hf_out = ask_hf(prompt)
        gm_out = ask_gm(prompt)

        time.sleep(5)

        results.append([
            name,
            q,
            get_num(hf_out),
            get_num(gm_out)
        ])
```

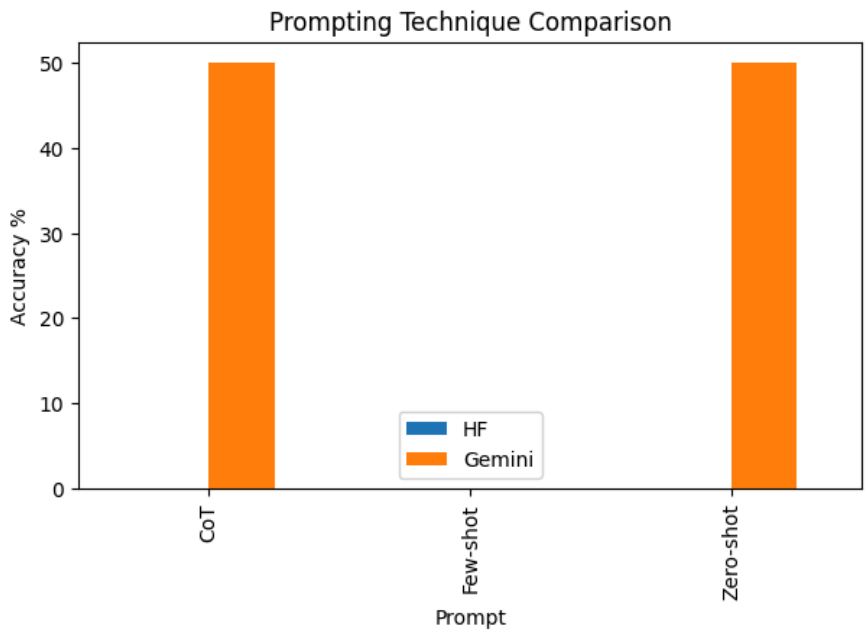
```python
# Table
df = pd.DataFrame(results, columns=["Prompt", "Question", "HF Output",
"Gemini Output"])
print(df)
```

OUTPUT:

```
       Prompt                                          Question Gold HF_Pred
0  Zero-shot  Janet’s ducks lay 16 eggs per day. She eats th...   18       2
1   Few-shot  Janet’s ducks lay 16 eggs per day. She eats th...   18       5
2        CoT  Janet’s ducks lay 16 eggs per day. She eats th...   18       2
3  Zero-shot  A robe takes 2 bolts of blue fiber and half th...    3     1.5
4   Few-shot  A robe takes 2 bolts of blue fiber and half th...    3      -4
5        CoT  A robe takes 2 bolts of blue fiber and half th...    3     1.5

  Gemini_Pred HF_Result  Gemini_Result
0        None     Wrong  Hallucination
1        None     Wrong  Hallucination
2        None     Wrong  Hallucination
3           3     Wrong        Correct
4        None     Wrong  Hallucination
5           3     Wrong        Correct

Accuracy (%)
            HF  Gemini
Prompt
CoT        0.0    50.0
Few-shot   0.0     0.0
Zero-shot  0.0    50.0
```



HF Summary
HF_Result
Wrong 6
Name: count, dtype: int64

Gemini Summary
Gemini_Result
Hallucination 4
Correct 2
Name: count, dtype: int64